Which Exposure Matters?

How training examples shape image-generation skills

Research project presentation

Proposal and exploratory work

We investigate whether image models require full combinations during training, or if learning constituent parts is sufficient. | Research in progress

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Adjacent research in generative vision

- **MOSAIC (Jeong et al., 2026):** Shows scene complexity drives failure more than raw concept imbalance, with composition collapsing under held-out combinations.
- **Constituent abilities (Okawa et al., 2023):** Demonstrates compositional skills emerge multiplicatively from individual constituent accuracies.
- **Optimization conflicts (GRASP, 2025):** Identifies gradient conflicts between head and tail classes during long-tail diffusion fine-tuning.

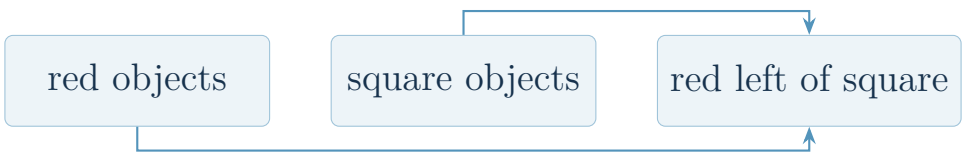
We propose to test which exposure level (predicate, pair, or full rule) is associated with generation success. Novelty requires further literature review.

Proposed contributions

- **Marginal-preserving manipulation:** Aim to vary joint rule exposure while matching individual exposures.
- **Falsifiable composition gap:** Compare success with a product reference on logit or log scales.
- **Threshold sample efficiency (N_{50}):** Traces parametric response trajectories rather than benchmarking arbitrary single checkpoints.
- **Constrained design optimization:** Explore allocations that improve exposure separation; feasibility remains to be established.

The question in one sentence

Does an image model need to see the **whole combination**, or are its parts enough?

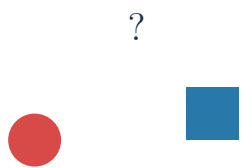


Can models combine separate concepts like red objects and squares into new spatial arrangements without explicit joint training examples?

Why should anyone care?

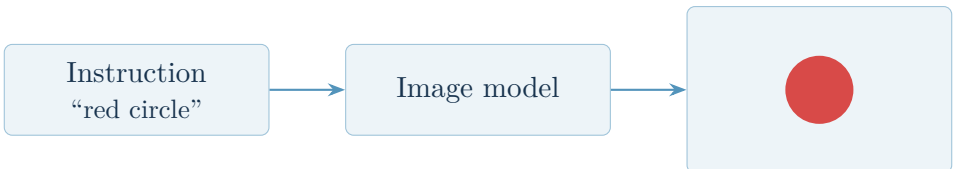
A model may recognize red items and blue squares separately, yet fail when asked to generate their spatial arrangement.

Understanding this failure helps us design more efficient training sets and diagnose compositional generalization limits.



Two familiar concepts

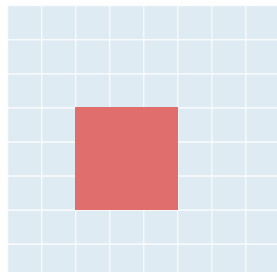
First: what is an image-generation model?



A conditional model takes an instruction and generates matching pixel arrays, learning visual patterns and statistical dependencies directly from observed training data.

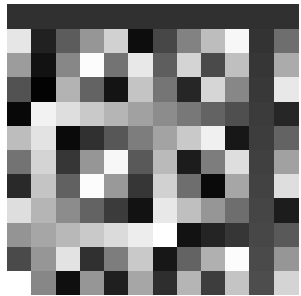
What is an image made of?

Digital images are rectangular grids of discrete pixels. Each pixel stores color intensities, so model inputs and outputs are large numeric arrays rather than abstract concepts.

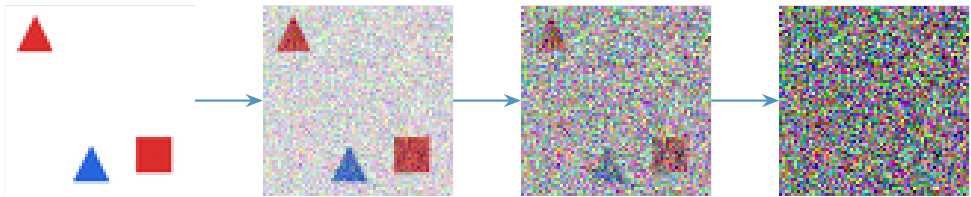


What is noise?

Noise consists of random variations in pixel values. As corruption increases, underlying shapes degrade until the image becomes pure static containing no recognizable visual information.



Diffusion: learn to reverse noise

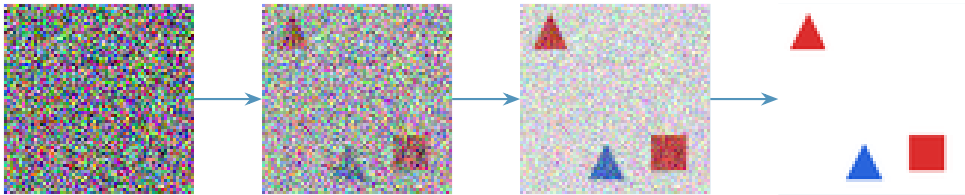


add noise

Forward diffusion progressively corrupts clean images with Gaussian noise. The network learns a prediction that helps reverse the corruption.

Illustrative noising, not the model's exact diffusion schedule.

Generation goes in the other direction



many denoising steps

Generation begins with pure random noise and repeatedly removes predicted noise over many steps, gradually transforming static into a sharp, recognizable visual scene.

Conceptual illustration; actual samples are generated by the model.

Forward diffusion: from image to noise

The forward diffusion process systematically corrupts a clean image x_0 across discrete timesteps $t \in \{1, \ldots, T\}$ via a Gaussian Markov chain:

$$q(x_t \mid x_{t-1}) = \mathcal{N}\left(x_t;\ \sqrt{1-\beta_t}\, x_{t-1},\ \beta_t \mathbf{I}\right)$$

Using cumulative variance $\bar{\alpha}_t = \prod_{s=1}^{t}(1-\beta_s)$, we sample x_t directly in closed form:

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \epsilon, \quad \epsilon \sim \mathcal{N}(0, \mathbf{I})$$

At timestep T, the signal is greatly reduced; the terminal distribution only approximates Gaussian noise.

The reverse process: noise prediction

Generative synthesis reverses corruption by learning parameterized Gaussian transitions:

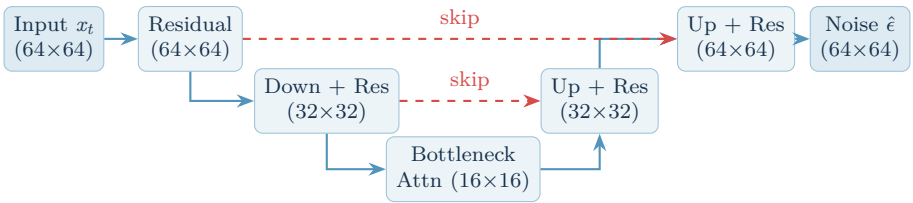
$$p_\theta(x_{t-1} \mid x_t, c) = \mathcal{N}\left(x_{t-1};\, \mu_\theta(x_t, t, c),\, \sigma_t^2 \mathbf{I}\right)$$

Reparameterizing the mean into noise estimation yields the simple MSE objective:

$$\mathcal{L}(\theta) = \mathbb{E}_{t, x_0, \epsilon, c}\left[\|\epsilon - \epsilon_\theta(x_t, t, c)\|^2\right]$$

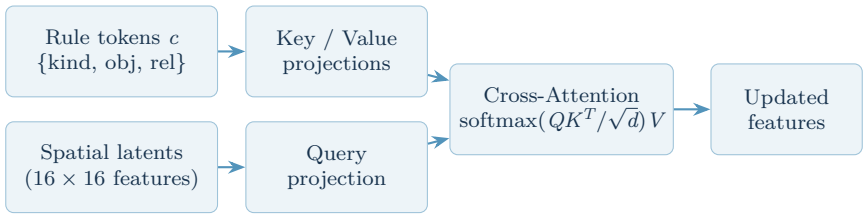
This is one common noise-prediction objective. Our later pilot uses velocity prediction.

Backbone architecture: hierarchical U-Net



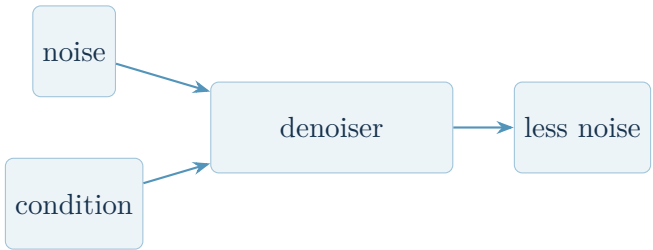
The U-Net combines multi-resolution downsampling and upsampling with skip connections, preserving fine spatial details while capturing global context.

Conditioning via cross-attention



Multi-head cross-attention at the 16×16 bottleneck binds symbolic predicate tokens directly to spatial feature maps, guiding object placement and colors.

What does “conditional” mean?



A conditional diffusion model receives both noisy pixels and an external instruction, guiding each denoising step toward images matching the prompt rather than arbitrary scenes.

Steering generation: classifier-free guidance

Classifier-Free Guidance (CFG) trains conditional and unconditional modes simultaneously by randomly dropping rule tokens ($p_{\text{drop}} = 10\%$) to a learned null embedding $\emptyset$.

During inference, predictions extrapolate along the condition vector:

$$\tilde{\epsilon}_\theta(x_t, c) = \epsilon_\theta(x_t, \emptyset) + s \cdot (\epsilon_\theta(x_t, c) - \epsilon_\theta(x_t, \emptyset))$$

Setting guidance scale $s > 1$ strengthens adherence to prompt relationships at the expense of sample diversity.

Training and generation are different

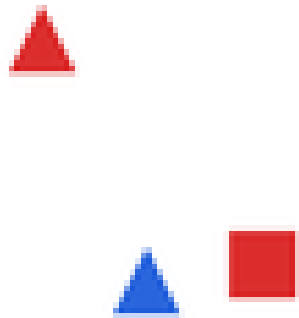
Training
Clean image + noise
↓
Predict denoising target

Generation
Random noise + prompt
↓
Repeated inference

Training uses known images and controlled noise to teach the denoiser; generation begins with pure noise and relies entirely on prompt guidance.

The scene world we control

- Synthetic 2D geometric scenes
- Verifiable bounding boxes and attributes
- Unambiguous concept exposure counting



Example generated by the project renderer.

We use a controlled synthetic world so concept frequencies and spatial arrangements can be measured with exact mathematical precision.

Our basic visual vocabulary

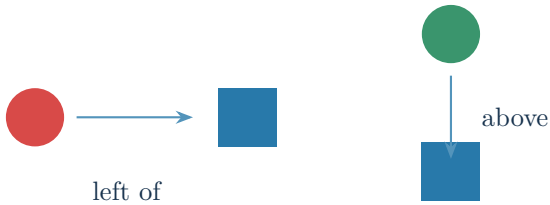
Colors
Red Blue
Green Yellow

Shapes
Circle Square
Triangle

The prototype features four colors and three shapes. Restricting visual primitives isolates core compositional binding mechanisms from natural image complexity.

Current prototype vocabulary; the final benchmark is not yet fixed.

Relationships add structure



Spatial predicates compare bounding box centers: left requires a smaller horizontal coordinate, while above requires a smaller vertical coordinate across distinct objects.

A rule is a combination of predicates

Example: **a red object left of a square**.



A composite rule binds multiple predicates together. Each predicate imposes an individual condition (color, shape, or relative position) that must be satisfied simultaneously.

Two roles mean two distinct objects



Valid: two red objects



Not enough: one red square

Relational rules specify two object roles that must bind to distinct objects. The two objects may share colors, but one object cannot fulfill both roles.

What do we mean by exposure?

Exposure is how often a pattern appears in the model's training scenes.

Exposure is the empirical fraction of training scenes satisfying an event. We compute exposure from verified ground-truth scene annotations, not just prompt requests.

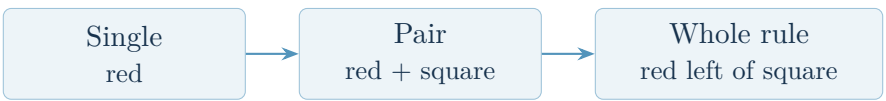
One scene can teach several things



- A red object appears
- A square appears
- Red is left of a square

A single training scene exposes the model to multiple concepts simultaneously: single attributes, pairwise combinations, and complete spatial relationships.

Three levels of exposure



We evaluate three exposure levels: single features, pairwise co-occurrences, and full rules. Our question is which level best predicts compositional success.

A simple counting example

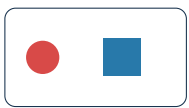


10 training scenes

When 3 of 10 sampled scenes satisfy a rule, its empirical exposure is 30%. In real datasets, finite sampling creates uncertainty that requires statistical estimation.

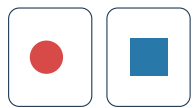
Why single-concept counts are not enough

Dataset A



Concepts appear together.

Dataset B



Concepts appear separately.

Both datasets can share identical individual concept frequencies, yet differ completely in joint co-occurrence. Marginal counts alone hide these joint exposure differences.

Lift: is a combination common or rare?

Lift compares observed joint exposure against an independent product reference:

$$\text{lift} = \log\left(\frac{\text{observed combination exposure}}{\text{product of individual exposures}}\right)$$

- Positive lift: concepts co-occur more often than expected by chance
- Negative lift: concepts co-occur less often than expected

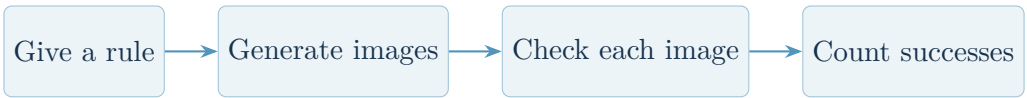
The product baseline is a reference; shared objects induce dependencies.

Illustrating lift with numbers



If red appears in 50% of scenes and square in 40%, independence predicts a 20% baseline. An observed 10% joint rate yields negative lift.

How do we measure model success?



$$\text{Observed success} = \frac{\text{classified images satisfying the rule}}{\text{total classified images}}$$

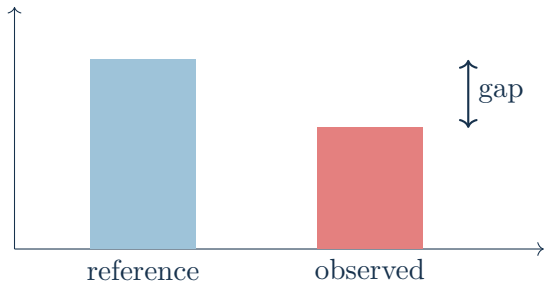
We evaluate generated images with an automated checker, reporting success rates while tracking and reporting checker abstentions separately from model generation errors.

A reference for composition

If constituent abilities combine independently, expected joint success equals the product of individual accuracies (for instance, $0.8 \times 0.7 = 0.56$).

Because shared objects induce dependencies, this calculation provides a testable empirical reference rather than an absolute theoretical performance ceiling.

The composition gap



The composition gap measures the deficit between observed rule success and the product reference, quantifying the extra difficulty of binding concepts together.

Illustration only; bars do not represent measured project results.

What does a zero gap mean?

When observed success equals the product reference, the gap is **zero**.

A zero gap means observed success equals the chosen product reference. The proposal uses logit scale; working drafts also explore a log-scale version.

Our main hypotheses

- **Complexity scaling:** Multi-predicate rules suffer larger negative composition gaps.
- **Joint exposure benefit:** Higher joint training exposure directly improves rule success.
- **Pairwise transfer:** Seeing pairs helps models generalize to unseen full combinations.

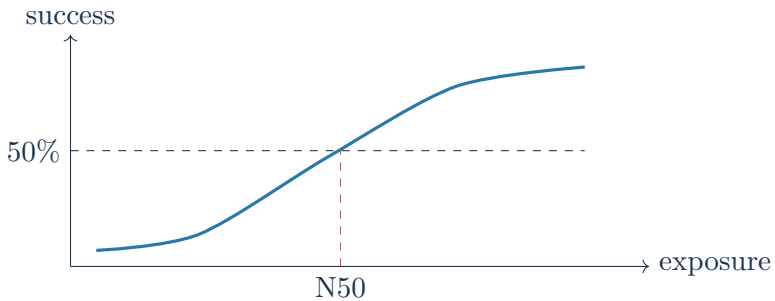
These hypotheses define the core empirical questions our study tests.

A separate question: how much is enough?

N50: an exposure level associated with 50% success.

Rather than evaluating a single arbitrary training checkpoint, we trace exposure-response curves. N50 marks where accuracy crosses 50%, measuring sample efficiency for rule learning.

An illustrative learning curve



Conceptual learning curves illustrate the transition toward mastery. Estimating N50 requires training allocations that span above and below the threshold without unsafe extrapolation.

Illustration only; not a fitted result.

Model optimization: AdamW and EMA

Weights are optimized by minimizing mean squared error with **AdamW**:

$$\theta_{k+1} = \theta_k - \eta_k \frac{\hat{m}_k}{\sqrt{\hat{v}_k} + \epsilon_{\text{opt}}} - \lambda \eta_k \theta_k$$

To smooth parameter updates, an **Exponential Moving Average (EMA)** shadow model is maintained:

$$\theta_{\text{EMA}} \leftarrow \mu\, \theta_{\text{EMA}} + (1 - \mu)\, \theta_k \quad (\mu = 0.999)$$

The pilot sampler uses EMA weights; better perceptual quality is not guaranteed.

Inference optimization: sampling schedules

DDPM (Stochastic Markovian)

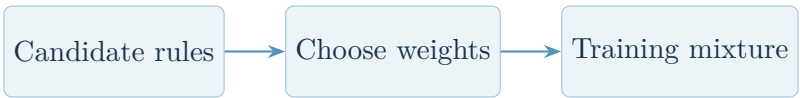
- Ancestral reverse sampling
- Injects Gaussian noise at each step
- Requires 100–1000 sequential passes

DDIM (Deterministic Non-Markov)

- Continuous ODE trajectory
- Deterministic mapping from noise
- Accelerates inference to 20–50 steps

DDIM and higher-order ODE solvers dramatically accelerate inference throughput without sacrificing visual fidelity or rule satisfaction.

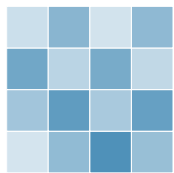
How do we change exposure fairly?



To isolate joint exposure, we adjust training mixture probabilities. This shifts how often whole rules appear while holding individual concept frequencies closely matched across datasets.

The exposure matrix

The exposure matrix maps requested conditions (columns) to realized visual events (rows). Cell entries are estimated by sampling ground-truth annotations rather than guessing from prompt labels alone.



Illustrative matrix; actual values estimated from sampled annotations.

Exposure design as constrained optimization

Finding balanced training mixtures is framed as convex optimization over condition weights $w \in \Delta^{K-1}$:

$$\min_{w} \; D_{\mathrm{KL}}(w \,\|\, u) = \sum_{k=1}^{K} w_k \log(K w_k)$$

subject to equality and inequality exposure constraints:

$$\mathbf{M}_{\text{marginal}}\, w = b_{\text{matched}}, \quad l \le \mathbf{M}_{\text{joint}}\, w \le u, \quad \sum w_k = 1, \; w_k \ge 0$$

Solving this via SLSQP produces the maximally uniform mixture that cleanly separates joint exposure.

A controlled comparison

Mixture A

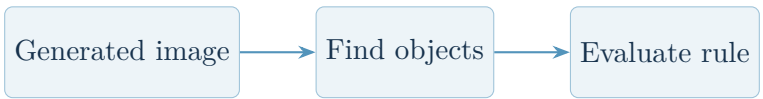
- Concepts matched
- Lower joint exposure

Mixture B

- Concepts matched
- Higher joint exposure

Comparing models trained on Mixtures A and B reveals whether joint co-occurrence drives compositional learning when individual concept frequencies remain strictly matched.

Why do we need a checker?



An automated checker detects objects and tests spatial relations from pixels. This provides repeatable, standardized evaluation across thousands of generated images without subjective human grading.

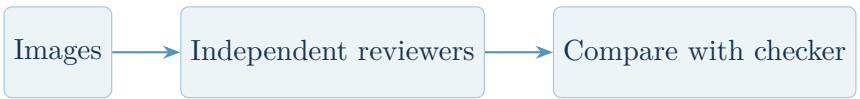
The checker can also be wrong

False accept
Checker says yes, but the image violates the rule.

False reject
Checker says no, but the image satisfies the rule.

When ambiguity prevents classification, the checker records an **abstention**. Abstentions must be tracked separately rather than treated as generation failures.

Humans help validate the checker

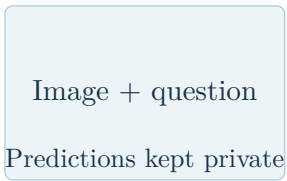


Before trusting the checker, human reviewers audit at least 300 complexity-stratified images. Comparing verdicts against the checker measures false accepts, false rejects, and per-predicate errors.

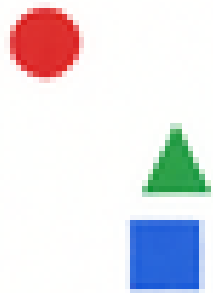
Why blind the review?

Reviewers inspect synthesized images alongside target questions without seeing automated checker predictions or training labels.

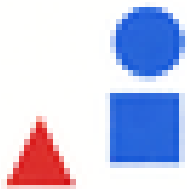
Blinding reduces the risk that checker predictions influence human judgments; it does not guarantee reviewer independence or perfect labels.



What does the current model produce?



Requested: red



Requested: blue above circle

Exploratory samples from the 800,000-update pilot show basic color generation, but multi-object relations show artifacts. These illustrate current prototypes rather than measured benchmark accuracy.

What have we built?

- Synthetic scene renderer and symbolic rule evaluator
- Exposure measurement and mixture optimization tools
- Pilot diffusion checkpoints, checkers, and review interfaces

Software components are operational, but completing infrastructure is distinct from validating the benchmark or running the full experiment.

What has the team reviewed?

- Conducted blinded calibration trials on small sample sets
- Discussed edge-case disagreements to refine rubrics
- Recorded and reported checker abstentions transparently

A 300-image, four-condition pilot audit pack is prepared; however, full complexity-stratified benchmark coverage remains outstanding.

One current obstacle: checker coverage

Some recognizable images are rejected by strict pixel and geometry checks.

The prototype checker enforces strict color and geometric heuristics, causing false rejections. More flexible thresholds improve recall but require independent validation against false accepts.

One current obstacle: relation exposure

An unrestricted relation can be almost always true in a three-object scene.

When spatial relations are tested without attribute constraints, multiple objects make relations like 'left of' almost trivially true. Full attributed rules represent completely different events.

One current obstacle: experimental design

We need designs that vary combinations while keeping other exposures controlled.

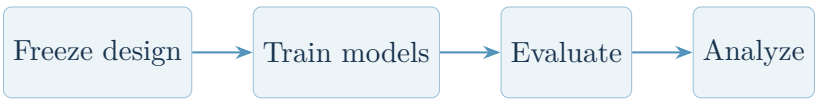
Current exploratory allocations fail the proposal's decorrelation targets and variance inflation criteria. Refined mathematical optimization is required to prevent confounding concept drift.

Stage 0: checks before the main study

- Human audit validation of automated checkers
- Exposure decorrelation and design stability
- Baseline comparison against adapted MOSAIC protocols

Current status: not passed. Main experimental model training will proceed only after all three Stage 0 validation gates are met.

What happens after the gates?



Once validation gates pass, we will freeze benchmark specifications, train controlled diffusion models, evaluate generated images with checkers, and analyze resulting composition gaps.

Planned workflow, not completed research findings.

Possible outcome: joint exposure matters

Heightened joint exposure could significantly improve multi-concept generation, even when individual concept frequencies are perfectly matched.

This positive finding would demonstrate that compositional binding requires dedicated co-occurrence examples during training, guiding future dataset construction.

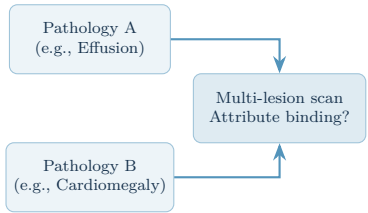
Possible outcome: the hypothesis fails

Increased joint exposure might yield minimal benefit, or effects could vary across rule types.

A negative result would be equally informative, suggesting that generative models generalize spatial arrangements compositionally from constituent exposure without needing explicit joint pairings.

Real-world implications: medical imaging

- **Rare co-morbidities:** Clinical scans frequently show individual pathologies, but dual conditions (e.g., effusion + cardiomegaly) are exceptionally sparse.
- **Diagnostic safety:** Synthetic augmentation requires verified attribute binding to avoid hallucinating false organ-lesion couplings.
- **Exposure thresholds:** Measuring N_{50} isolates the minimal co-occurrence needed for clinically faithful scan synthesis.



Controlled exposure reveals whether clinical models generalize to rare joint diseases.

Real-world implications: robotics and safety

- **Autonomous vehicles:** Rare safety-critical corner cases combine familiar entities in unseen configurations (e.g., pedestrian behind an open vehicle door in fog).
- **Robotic manipulation:** Goal-image generation for robot policies demands strict geometric binding between grasp targets and surrounding obstacles.
- **Defect simulation:** Industrial inspection models must simulate multi-defect interactions from sparse marginal failure observations.

Our exposure-control framework provides the mathematical foundation to audit whether generative simulators reliably synthesize hazardous edge cases.

Why this research matters

- Clarifies whether generative models require joint observations
- Establishes rigorous methods to measure realized training exposure
- Guides principled allocation of compute and dataset collection budgets

Empirical measurement replaces guesswork in training set design for compositional image generation.

Take-home message

A model's success may depend on **which combinations** it has seen, not only how often it has seen the pieces.

By controlling synthetic environments, tracking true exposures, and auditing checkers, our project tests whether joint training examples drive compositional skill.

Thank you

Core question: Which training examples help a model combine concepts?

Open for questions regarding synthetic controls, exposure metrics, checker validation, and next experimental steps.

References and project evidence

- *Which Exposure Matters?* Research proposal, revision 4.
- Jeong et al., MOSAIC, ICML 2026: controlled scene complexity and imbalance.
- Okawa et al., NeurIPS 2023: compositional abilities and constituent abilities.
- Nützel et al., GRASP, arXiv 2025: gradient conflicts in long-tail diffusion.
- Bradley, ICML 2026: local conditional scores in compositional diffusion.
- Project documentation: stage status, checker, human audit and exposure design.

Prior-work descriptions follow the proposal; consult the original publications for details.